

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI



Project Phase -1 Synopsis
on

**“AI-Based Coding Reward Platform with a
Goal-Oriented Wallet and Controlled Withdrawal
System”**

submitted in partial fulfillment of the requirements for the project phase - 1 of degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

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Institute Vision

To be the most preferred institution for engineering & management education, research and entrepreneurship by creating professionally superior and ethically strong global manpower.

Institute Mission

To prepare students for professional accomplishments, to provide education through the committed, highly skilled faculty by partnering and collaborating with industry and R&D institutes.

Department Vision

To be a leader in providing education with skilled technical knowledge and professional ethics, creating competent and ethically strong IT professionals.

Department Mission

- We impart quality education to students by ensuring a learning environment through qualified faculty and a good infrastructure.
- We can encourage students to attain technical and ethical skills for a successful career in industry, academics, research & entrepreneurship through active engagement with all the stakeholders.

Program Educational Objectives (PEOs)

Computer Science and Engineering Graduates within 3-5 years of Graduation should have-

PEO 1: The ability to conceptualise, analyse, design and develop IT Solutions of varying complexities by leveraging advances in computer technology.

PEO 2: The ability to apply standard practices and strategies in software project development and management using an industry-wide benchmarked framework to deliver a sustainable quality product.

PEO 3: The ability to work as a team player in a cross-cultural environment adhering to work ethics with a passion for entrepreneurship and a zest for higher studies.

Program Specific Outcomes (PSOs)

The Computer Science and Engineering Graduates Will Have the Ability to-

PSO 1: To develop the ability to interpret the fundamental concepts and methodologies of computer systems.

PSO 2: To use mathematical concepts to crack problems using suitable mathematical analysis, data structures and algorithms.

PSO 3: Develop the ability to grasp the software development lifecycle and methodologies of software systems and possess competent skills and knowledge of the software design process. Familiarity and practical proficiency with a broad area of programming concepts and provide new ideas and innovations towards research.

Abstract

Programming skill development is essential for academic success and career growth, yet existing coding platforms focus primarily on rankings and competition rather than structured financial empowerment. This project proposes an **AI-Based Coding Reward Platform with a Goal-Oriented Wallet and Controlled Withdrawal System** to transform coding practice into measurable financial goal achievement. The system enables users to set specific financial objectives, such as purchasing a laptop or funding a course, and earn reward points by solving coding problems of varying difficulty levels. A machine learning-based evaluation engine analyzes correctness, efficiency, and complexity of submitted solutions to assign dynamic performance-based rewards. Accumulated points are securely stored in a goal-oriented digital wallet, which tracks progress toward predefined financial targets. A controlled withdrawal mechanism incorporating authentication and fraud detection ensures secure reward redemption. Additionally, a goal-based advertisement module supports platform sustainability through contextual monetization. The proposed system aims to provide a secure, scalable, and intelligent ecosystem that integrates AI-driven evaluation, gamification, financial tracking, and secure transaction management to promote both technical skill enhancement and structured financial discipline.

The project contributes to Sustainable Development Goal (SDG) 4: Quality Education by encouraging regular coding practice, SDG 8: Decent Work and Economic Growth by linking skill development with financial rewards, and SDG 9: Industry, Innovation, and Infrastructure through the integration of artificial intelligence and digital finance systems.

Synopsis on “AI-Based Coding Reward Platform with a Goal-Oriented Wallet and Controlled Withdrawal System”

1 Introduction

In today’s digital world, programming skills are essential for academic success and career growth. Many students regularly practice coding on online platforms, but they often lack financial motivation and structured systems that connect skill development with personal financial goals. Existing coding platforms mainly focus on ranking, competition, and certificates rather than helping students achieve real-life financial objectives. To address this gap, this project proposes an AI-Based Coding Reward Platform with a Goal-Oriented Wallet and Controlled Withdrawal System.

The platform allows users to set a specific financial goal such as purchasing a laptop, paying course fees, or saving for travel. Users earn reward points by solving coding problems. Artificial Intelligence evaluates their performance based on correctness, difficulty level, and efficiency of the solution. The earned points are stored in a digital wallet linked to the selected goal. Once the required points are achieved, users can request a controlled withdrawal. The system also uses goal-based advertisement logic and sponsored coding challenges to generate revenue for maintaining platform sustainability. This project transforms coding practice into structured financial growth while ensuring fairness, transparency, and security.

2 Literature Survey

Investigates the role of gamification in enhancing collaborative learning among students. The study identifies a major limitation in traditional education systems, where individual performance is emphasized while teamwork and collaboration are often neglected. To address this issue, the authors implemented the Classcraft gamification platform, which integrates game elements such as points, rewards, and team-based challenges into classroom activities. The research employed a quantitative experimental methodol-

ogy involving pre-test and post-test evaluations to measure improvements in collaborative skills. The findings demonstrate a significant enhancement in students' teamwork abilities, including improved communication, group coordination, and shared responsibility. The use of gamification increased student engagement and encouraged active participation in learning tasks. However, the study primarily focuses on collaborative learning outcomes rather than individual academic performance or programming skill development. Additionally, the system lacks automated evaluation mechanisms and does not incorporate AI-based feedback to assess individual contributions. This limitation reduces its applicability in technical domains such as programming education. Nevertheless, the paper strongly supports the effectiveness of gamification in improving student engagement and learning behavior. It is highly relevant to the proposed project as it validates the use of gamification techniques, but it also highlights the absence of intelligent evaluation and structured reward systems, which are addressed in the proposed AI-based coding reward platform. [1]

Provides a comprehensive overview of how machine learning techniques are applied to measure similarity between source code. The authors analyze a large number of research studies and identify various applications of code similarity, including plagiarism detection, code clone detection, and software quality analysis. The study highlights that traditional methods of code comparison, such as syntactic and structural analysis, are limited in handling complex code variations, whereas machine learning approaches offer more accurate and scalable solutions. Different ML techniques such as supervised learning, unsupervised learning, and deep learning models are discussed, along with their effectiveness in identifying patterns and similarities in code. The paper also examines commonly used representations like abstract syntax trees (AST) and datasets such as BigCloneBench, which are widely adopted for evaluation. Furthermore, it identifies key challenges in the field, including lack of standardized datasets, computational complexity, and difficulty in handling semantic similarity. Although the paper provides a broad and detailed analysis of existing methods, it mainly focuses on technical aspects of code similarity and does not directly address student learning, gamification, or reward-based systems. Overall, this study is significant as it supports the use of AI and machine learning for automated code evaluation, which is highly relevant to developing intelligent programming learning platforms. [2]

Explores the effectiveness of discovery-based learning approaches in enhancing programming education. The study addresses the common challenges faced by novice programmers, such as difficulty in understanding programming logic, lack of problem-solving skills, and dependency on instructor guidance. To overcome these challenges, the authors propose the use of WebQuest-based discovery learning, where students actively explore online resources, analyze information, and solve programming problems independently. The research findings indicate that this approach significantly improves students' analytical thinking, creativity, and ability to apply programming concepts in real-world scenarios. Students demonstrated better engagement and motivation compared to traditional teaching methods. However, the study does not incorporate automated evaluation systems or intelligent feedback mechanisms to assess student performance. Additionally, it lacks gamification elements and reward-based motivation strategies, which could further enhance student engagement. The absence of AI-based feedback limits the system's ability to provide personalized learning experiences. This paper is relevant to the proposed project as it highlights the importance of active learning and student-centered approaches in programming education. However, it also reveals the need for integrating advanced technologies such as AI-based evaluation, gamification, and reward systems to create a more comprehensive learning platform, which is the objective of the proposed system. [3]

Examines the effectiveness of gamification in teaching complex technical subjects such as blockchain. The study introduces an experiential learning framework that integrates gamification elements to enhance students' cognitive engagement and learning outcomes. The proposed system allows students to actively participate in learning cycles involving experience, reflection, and experimentation, which helps them understand complex concepts more effectively. The results of the study indicate that gamification significantly improves knowledge retention, cognitive well-being, and student motivation. Students showed increased interest in learning and were able to grasp difficult concepts more easily. However, the system primarily focuses on conceptual understanding rather than evaluating programming skills or performance. It does not include automated coding assessment or AI-based feedback mechanisms. Additionally, the reward system is limited to gamification elements and does not incorporate structured or secure reward management. This paper is relevant to the proposed project as it demonstrates the effectiveness

of gamification in technical education. However, it highlights the need for integrating intelligent evaluation and reward systems, which are key components of the proposed AI-based coding reward platform. [4]

Presents a comprehensive gamification framework designed to enhance student engagement in STEM education. The authors propose the NEWTON-Enhanced Gamification Model (N-EGM), which integrates multiple gamification layers such as points, rewards, goals, and social interaction into a learning management system. The framework addresses several challenges in traditional gamification approaches, including lack of personalization and limited integration with educational technologies. The system was implemented and evaluated in real educational settings, showing significant improvements in student engagement, participation, and knowledge acquisition. The study emphasizes the importance of embedding gamification throughout the learning process rather than treating it as an add-on feature. However, the framework does not incorporate AI-based coding evaluation or automated performance assessment. Additionally, it lacks a structured reward management system, such as a digital wallet, to manage incentives. While the paper provides a strong foundation for designing gamified learning systems, it highlights the need for integrating intelligent evaluation and secure reward mechanisms. This gap is addressed in the proposed project, which aims to combine gamification with AI-based coding evaluation and reward management. [5]

A Study of Engagement, Motivation, and Performance” published in IEEE Access examines the effectiveness of integrating gamification and game-based learning into web development education. The study addresses the challenges students face in learning complex web technologies such as HTML, CSS, and programming concepts, which often lead to reduced engagement and motivation. To evaluate the impact, the authors conducted an experimental study involving 60 university students, where one group used a gamified learning system while the control group followed traditional teaching methods. The gamified system was designed based on Self-Determination Theory and Flow Theory to ensure balanced learning experiences and intrinsic motivation. The results showed that students using the gamified system achieved better performance compared to the non-gamified group. However, the improvement in engagement and motivation was not consistently significant across all students, indicating that gamification effectiveness depends on individual learner differences. The study also highlights the “novelty effect,”

where initial interest in gamification may decrease over time if not properly sustained. While the paper strongly supports the use of gamification in programming education, it does not include AI-based evaluation or personalized feedback mechanisms. Additionally, the reward system is limited to basic gamification elements and does not include structured or secure reward management. This paper is highly relevant to the proposed project as it validates the effectiveness of gamification in coding education, but it also emphasizes the need for integrating intelligent evaluation and reward systems, which are addressed in the proposed platform. [6]

Provides a comprehensive analysis of automated feedback mechanisms used in programming education. The study highlights the importance of timely and personalized feedback in helping students understand programming concepts and improve their skills. It reviews various feedback approaches, including static analysis, unit testing, machine learning models, and large language model-based systems . The findings indicate that automated feedback systems significantly enhance learning outcomes by providing immediate and consistent evaluation, especially in large-scale educational environments such as MOOCs. The study also reveals that most existing systems focus on identifying errors rather than guiding students toward correct solutions. Furthermore, while AI-based feedback systems are becoming more advanced, many lack personalization and adaptability to individual learning needs. Another limitation is the absence of integration with gamification and reward mechanisms, which could further motivate students. This paper is highly relevant to the proposed project as it supports the use of AI-based evaluation in programming education. However, it highlights the need to combine feedback systems with gamification and incentive mechanisms to create a more engaging and effective learning platform, which is the goal of the proposed system. [7]

Presents a systematic literature review on the application of personalized gamification in educational systems. The study emphasizes that traditional gamification approaches often fail to consider individual differences among learners, leading to reduced effectiveness. To address this, the authors explore personalized gamification techniques that adapt game elements such as points, rewards, and challenges based on individual learning styles and preferences . The findings indicate that personalized gamification significantly improves student motivation, engagement, and learning outcomes compared to generic gamification systems. The study also highlights the integration of advanced technologies

such as artificial intelligence, augmented reality, and mobile learning platforms in implementing personalized gamification. However, the paper mainly focuses on theoretical analysis and does not provide a fully implemented system. Additionally, it lacks integration with automated coding evaluation and structured reward management systems. This paper is relevant to the proposed project as it highlights the importance of personalization in gamified learning environments. However, it also indicates the need for combining personalized gamification with AI-based evaluation and secure reward systems, which are key components of the proposed platform. [8]

Investigates the impact of gamification on student engagement and learning outcomes in programming education. The study focuses on the use of gamification elements such as points, badges, and leaderboards to encourage participation and motivation among students learning the C programming language. The research employed a mixed-method approach, including surveys, system logs, and performance evaluations to analyze student behavior and learning outcomes. The results show that gamification significantly increases student engagement and participation in learning activities. Students reported higher motivation and interest in completing tasks, and there was a moderate improvement in learning outcomes. However, the study also indicates that the impact of gamification on academic performance is not always significant. Additionally, the system does not include AI-based evaluation or automated feedback mechanisms, limiting its ability to provide personalized learning experiences. The reward system is also limited to basic gamification elements and does not include structured or secure reward management. This paper is relevant to the proposed project as it supports the role of gamification in improving student engagement. However, it highlights the need for integrating intelligent evaluation and reward systems to achieve better learning outcomes. [9]

Explores the use of gamification to improve student awareness of academic integrity in programming education. The study introduces a gamified system that integrates plagiarism detection with game elements such as points, badges, and leaderboards to encourage students to submit original work and engage with feedback reports. The system was evaluated through multiple experiments involving undergraduate students, and the results indicate that gamification increases student engagement, encourages early submission of assignments, and improves awareness of plagiarism and collusion. Students were more likely to interact with feedback reports and improve their behavior when gamification

elements were included. However, the system primarily focuses on academic integrity rather than improving programming skills or performance. It does not include AI-based coding evaluation or personalized feedback mechanisms. Additionally, the reward system is limited to gamification elements and does not include structured or secure reward management. This paper is relevant to the proposed project as it demonstrates the effectiveness of gamification in influencing student behavior. However, it highlights the need for integrating gamification with AI-based evaluation and reward systems to create a comprehensive coding learning platform. [10]

Research Gap

The existing literature demonstrates that gamification techniques significantly improve student engagement and motivation, while AI-based learning systems provide effective support through automated evaluation and personalized feedback. Additionally, studies on incentive and reward mechanisms show that rewards can enhance user participation and learning behavior. However, most of these approaches are developed and studied independently, focusing on only one aspect of the learning process. There is a clear lack of a unified system that integrates AI-based coding evaluation, gamified learning environments, and structured reward management systems into a single platform.

Furthermore, many existing systems do not provide real-time personalized feedback that adapts to individual student performance. Gamification-based platforms often rely on basic elements such as points and badges but lack meaningful performance-based reward distribution. Similarly, reward and incentive systems are not effectively linked with actual learning outcomes. Another limitation is the absence of secure and goal-oriented reward management systems, such as digital wallets, to track and utilize earned rewards. Therefore, there is a need for a comprehensive solution that integrates AI-driven evaluation, gamification strategies, and secure reward systems to enhance both learning efficiency and student motivation, which is the objective of the proposed system.

Literature Survey Table

S.No	Author Name	Title of the Paper	Drawback
1	Maraza-Quispe et al.	Impact of Gamification on Collaborative Learning Development	Focuses on teamwork improvement but lacks AI-based evaluation and individual performance analysis.
2	Zixian Zhang et al.	Machine Learning Approaches to Code Similarity Measurement: A Systematic Review	Focuses on technical aspects of code similarity and does not include educational, gamification, or reward-based systems.
3	Aldalur et al.	Improving Programming Learning in Engineering Students Through Discovery Learning	Improves learning approach but lacks gamification and automated evaluation systems.
4	Tsang et al.	Gamified Blockchain Education in Experiential Learning	Focuses on conceptual learning, lacks coding evaluation and reward management system.
5	Zhao et al.	An Innovative Multi-Layer Gamification Framework for Improved STEM Learning Experience	Enhances engagement but lacks AI-based performance evaluation and secure reward system.
6	Ojonuba et al.	Enhancing Web Development Education With Game-Based and Gamification Learning	Improves performance but lacks personalized AI feedback and structured reward mechanisms.
7	Castellanos et al.	Feedback Systems on Computer Programming Courses: A Systematic Literature Review	Focuses on feedback but lacks gamification and reward-based motivation.

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S.No	Author Name	Title of the Paper	Drawback
8	Ristianto et al.	Personalized Gamification: A Technological Approach for Student Education	Focuses on personalization but lacks implementation with AI evaluation and reward systems.
9	Ibanez et al.	Gamification for Engaging Computer Science Students in Learning Activities	Improves engagement but lacks AI-based coding evaluation and structured reward system.
10	Karnalim et al.	Gamification to Help Inform Students About Programming Plagiarism and Collusion	Focuses on plagiarism awareness, not on coding skill improvement or reward management.

Conclusion

The reviewed literature highlights that gamification, AI-based learning systems, and incentive mechanisms play a significant role in improving student engagement and performance in programming education. Several studies demonstrate that gamification techniques such as points, badges, and leaderboards effectively enhance motivation and participation. Other works emphasize the importance of automated feedback and AI-based evaluation systems in providing timely and personalized learning support. Additionally, research on incentive mechanisms shows that reward-based systems can encourage active participation and improve user involvement.

However, most existing studies focus on individual components rather than providing a complete solution. There is limited work that integrates AI-based coding evaluation, gamification strategies, and structured reward management systems into a single platform. Therefore, the proposed system aims to combine these aspects to create a comprehensive and effective coding learning environment that improves both learning outcomes and student motivation.

3 Problem Statement

Although coding platforms have become popular for improving programming skills, they mainly emphasize problem-solving, coding contests, and rankings. Features such as AI-based performance evaluation, real monetary rewards, goal-oriented savings, and secure wallet management are not commonly integrated into a single platform. As a result, users may lose motivation to practice coding consistently. Therefore, there is a need to develop an AI-based coding reward platform that combines intelligent code evaluation with a reward-based wallet system to encourage continuous learning and skill development.

4 Objectives

The objectives of this project are as follows.

1. To develop an AI-based system that evaluates coding solutions and assigns reward points based on performance.
2. To design a goal-oriented digital wallet that stores earned points and tracks goal progress.
3. To implement a controlled withdrawal system that allows users to redeem rewards after completing their financial goal.
4. To display relevant advertisements based on the user's selected goal to support platform revenue.

5 Methodology

The system begins with user registration and selection of a financial goal, and based on this goal, relevant advertisements are displayed to the user. Users solve coding problems, and their solutions are checked for correctness. Artificial Intelligence evaluates the performance based on accuracy and difficulty and assigns reward points. The earned points are stored in a goal-oriented wallet that tracks progress toward the selected goal. Once the required points are achieved, the user can request secure withdrawal, as shown in Figure 2.

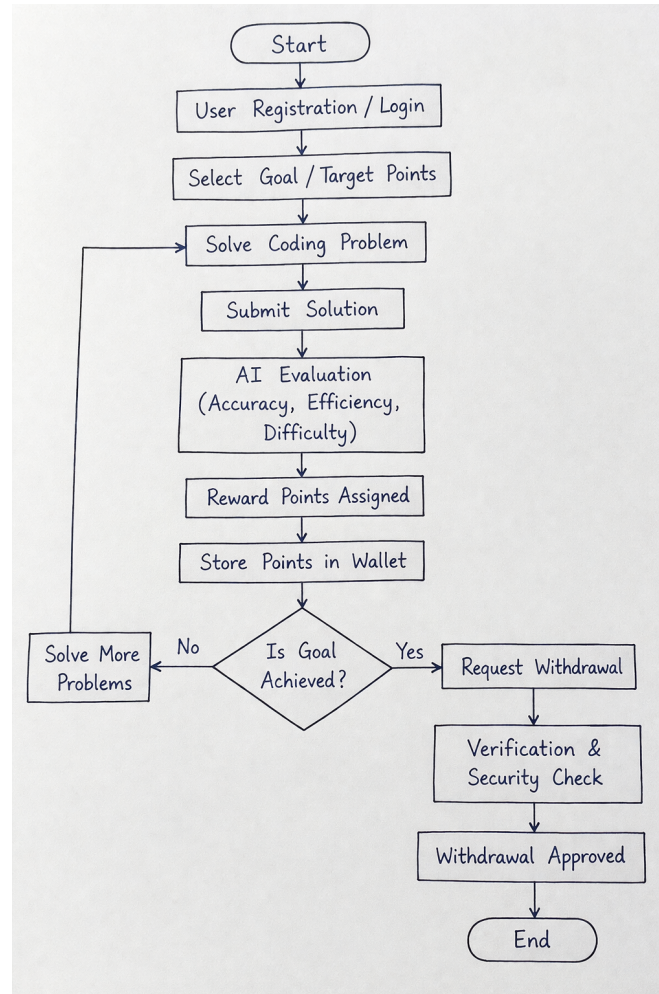


Figure 1: Methodology of AI-Based Coding Reward Platform

6 Technology

The proposed AI-Based Coding Reward Platform is developed using modern web technologies, cloud services, and AI tools. The frontend provides an interactive user interface, while the backend manages authentication, code execution, reward calculation, and wallet management. MongoDB Atlas is used as the cloud database, Judge0 API executes C programs, and Google Gemini API assists in AI-based code evaluation. The complete technology stack used in the project is presented in Table 2.

Table 2: Recommended Technology Stack

Layer	Recommended Stack	Purpose
Frontend	Next.js (React), Tailwind CSS, Monaco Editor	Provides a modern user interface with an integrated online code editor for C programming.
Backend	Node.js (Express.js)	Handles user authentication, APIs, reward calculation, wallet management, and communication with external services.
Database	MongoDB Atlas	Stores user details, coding problems, submissions, wallet balance, reward history, and transaction records.
Code Execution	Judge0 API	Compiles and executes C programs securely using predefined test cases.
AI Evaluation	Google Gemini API + Python	Evaluates code quality, logic, efficiency, and generates an intelligent performance score.
Advertisement	Google AdSense	Displays advertisements to generate revenue for the reward system.
Hosting	Vercel (Frontend), Render (Backend)	Provides free cloud hosting with secure HTTPS deployment.

7 System Architecture

The proposed AI-Based Coding Reward Platform follows a client-server architecture consisting of a frontend, backend, cloud database, AI evaluation module, code execution service, and reward management system. The user interacts with the platform through a Next.js-based frontend, which provides modules for registration, login, coding practice, wallet management, and goal tracking. The frontend communicates with the Node.js (Express.js) backend, which processes user requests, manages authentication, calculates rewards, and controls wallet operations.

User information, coding submissions, wallet balance, reward history, and withdrawal requests are securely stored in MongoDB Atlas, a cloud-based NoSQL database. When a user submits a C program, the backend sends the code to the Judge0 API, which compiles and executes the program against predefined test cases. The execution results are then analyzed using Google Gemini API along with Python-based evaluation logic to assess

code quality, correctness, and performance. Based on the AI evaluation, reward points are calculated and credited to the user's goal-oriented wallet. The platform generates revenue through Google AdSense, which supports the reward mechanism. Once the user's target goal is achieved, the accumulated rewards can be withdrawn through a secure withdrawal process. Figure 2.

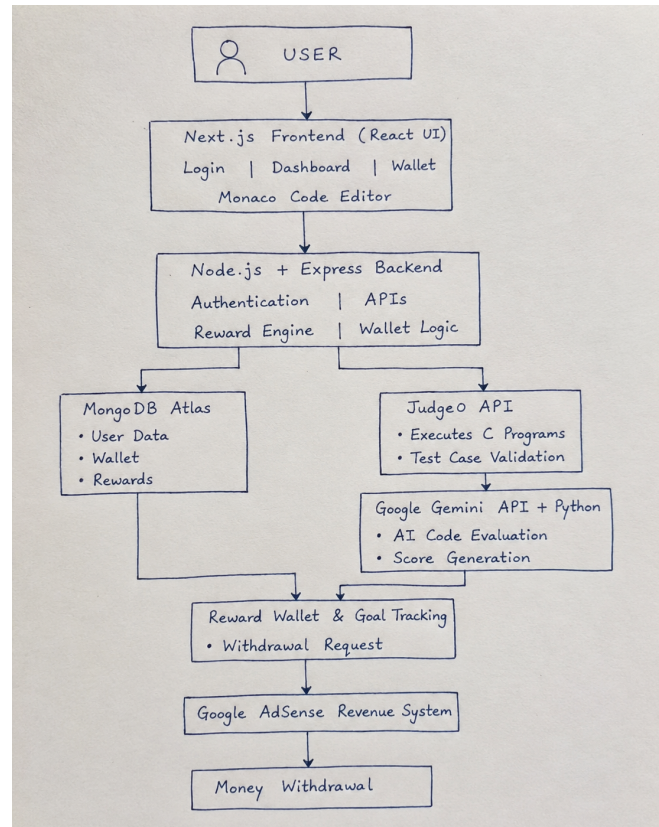


Figure 2: System Architecture of the AI-Based Coding Reward Platform

8 Implementation

The proposed AI-Based Coding Reward Platform will be implemented using a modular approach. Initially, the system will provide user registration and authentication, followed by a coding environment where users can solve C programming problems. The submitted programs will be executed against predefined test cases, and the execution results will be evaluated to determine the correctness and performance of the solution.

Based on the evaluation results, the reward engine will calculate reward points and credit them to the user's goal-oriented digital wallet. Users will be able to track their reward points and monitor their progress toward their selected financial goals through the

dashboard. The platform will also integrate an advertisement module to generate revenue that supports the reward mechanism. Finally, a secure withdrawal module will be implemented to allow users to request withdrawal of their earned rewards after successfully achieving their selected goals.

9 Scope

The scope of this project includes developing an AI-based coding evaluation engine, designing a secure digital wallet system, implementing a goal-tracking mechanism, and building a controlled withdrawal process. The project also includes integrating intelligent advertisement logic to maintain financial sustainability. In the future, the system can be expanded by adding mobile application support, internship integration, advanced AI analytics, and blockchain-based transparency for enhanced trust and scalability.

10 Purpose

The purpose of this project is to motivate students to improve their coding skills while helping them achieve personal financial goals in a structured manner. It aims to transform daily coding practice into measurable financial growth through intelligent evaluation and disciplined saving. The system promotes skill development, financial awareness, and technological innovation by combining AI with goal-based digital finance management.

11 Existing Work

Current coding platforms focus mainly on competitive programming, rankings, and certification systems. While they successfully enhance technical skills, they do not provide structured financial saving systems. Some freelancing platforms offer earning opportunities, but they require professional-level expertise and are not suitable for beginners. There is no widely adopted platform that integrates AI-based performance evaluation, goal-linked wallet systems, controlled withdrawal processes, and advertisement-driven sustainability in a single ecosystem. This project addresses that gap by combining education, artificial intelligence, and fintech concepts into one innovative platform.

12 Relevance to Sustainable Development Goals (SDGs)

This project aligns with the following UN Sustainable Development Goals (SDGs):

- **SDG 4 – Quality Education:** The platform promotes continuous learning and improves programming skills through regular practice.
- **SDG 8 – Decent Work and Economic Growth:** It supports skill-based earning opportunities and financial motivation for students.
- **SDG 9 – Industry, Innovation and Infrastructure:** The project encourages innovation by integrating artificial intelligence, digital wallets, and smart monetization systems.

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